

MADHAV KUMAR

Aspiring Software Engineer

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PROFILE

Final-year B.Tech Computer Science student with ongoing Software Development Engineer internship at Nielsen, specializing in AWS serverless architecture, Python/Java backend development, containerization, and CI/CD automation.

TECHNICAL SKILLS

Core Skills:

Python, Java, AWS (Lambda, S3, EC2, RDS, DynamoDB, SQS, SNS, CloudWatch, IAM), Docker, Kubernetes, Terraform, PostgreSQL, ClickHouse, GitLab CI/CD, Git

Additional Experience:

Spring Boot, MySQL, MongoDB, REST APIs, Microservices, Linux, Jira, Confluence

EXPERIENCE

Nielsen – Software Development Engineer Intern

Mar 2025 - Present

- Engineered an automation framework for AWS Lambda validation using Python and CloudWatch, eliminating repetitive log checks and reducing manual testing time by about 60 percent.
- Integrated the validation framework into GitLab CI/CD pipelines, enabling both on-demand manual execution across environments and automatic UAT validation on every Lambda deployment.
- Enhanced existing Lambda and Docker media workflow to compress WAV to FLAC (70 percent smaller), reducing S3 storage footprint across regions.
- Implemented PostgreSQL-driven asset evaluation logic and published structured events across SNS to SQS pipelines.
- Enhanced asset-tracking Lambdas to generate real-time SQS events and validated thousands of ClickHouse ingested records for accuracy.
- Designed an automated web-scraping framework for Amazon Prime Video metadata, running twice daily and collecting between 1,000 and 1,500 episode records per cycle to support internal data workflows.
- Analyzed large-scale S3 digital footprints through Python scripts to detect asset metadata gaps.
- Updated Lambda functions' runtimes and code using Java and Python, ensuring the system remains up-to-date and operates efficiently.
- Proactively acquired expertise in AWS (EC2, DynamoDB), Docker, GitLab CI/CD, and Kubernetes (deployments, services, ConfigMaps, Secrets, sidecars).

PROJECTS

GitForgeX – Custom Version Control System

- Built a Git-inspired Version Control System using locally running containerized services, including a Repository Management Service and a Version History Service.
- Developed secure handshake-based push operations, API-driven inter-service communication, and snapshot-based repository storage with low-latency handling of multiple requests.

Kubernetes Logging System (Personal Project)

- Built a Kubernetes-based sidecar system using ClickHouse and Fluentd for log collection and analysis.
- Successfully collected and queried over 50,000 log events in local tests with sub-second response times.

Sentiment Analysis API (Spring Boot + Java NLP)

- Developed a Spring Boot application that analyzes user input text using Stanford CoreNLP to detect sentiment.
- Designed a simple REST API for real-time sentiment classification and response delivery.

Task Scheduler App (Android – Kotlin + SQLite)

- Created a fully offline Android task manager with support for recurring tasks, categories, and instant search.
- Tested with 2,000+ tasks on mid-range devices while maintaining smooth performance and < 50 ms search response.

CERTIFICATES

Application Development in Java

- Gained practical knowledge in Java GUI development, JDBC for database management, and Java networking. Applied these skills in projects to improve application design and functionality.

EDUCATION

Lovely Professional University

Bachelor of Technology-Computer Science & Engineering - 87.60%

Jalandhar, Punjab, India
August 2022-June 2026

Litera Valley School

Senior secondary education (PCM) - 81%

Patna, Bihar, India
2019 - 2021

PTJM Saraswati Vidya Mandir

Secondary education (10th) - 92%

Rajgir, Bihar, India
2018 - 2019